IN THIS CHAPTER

» **Examining the basics of cloud computing**

» **Looking at three kinds of cloud computing services**

» **Understanding the pros and cons of cloud computing**

» **Perusing a few major cloud computing service providers**

Chapter 6
Cloud Computing

The world's two most popular science-fiction franchises — *Star Wars* and *Star Trek* — both feature cities that are suspended in the clouds. In *The Empire Strikes Back*, Han Solo takes the *Millennium Falcon* to Cloud City, hoping that his friend Lando Calrissian can help repair their damaged hyperdrive. And in the original *Star Trek* series episode "The Cloud Minders," the crew of the *Enterprise* visits a city named Stratos, which is suspended in the clouds.

Coincidence? Perhaps. Or maybe Gene Roddenberry and George Lucas both knew that the future would be in the clouds. At any rate, the future of computer networking is rapidly heading for the clouds — cloud computing, to be specific. This chapter is a brief introduction to cloud computing. You discover what it is, the pros and cons of adopting it, and what services are provided by the major cloud-computing providers.

Introducing Cloud Computing

The basic idea behind cloud computing is to outsource one or more of your networked computing resources to the internet. The cloud represents a newish way of handling common computer tasks. Table 6-1 outlines just a few examples of how the cloud way differs from the traditional way.

TABLE 6-1 Traditional versus Cloud Computing

	Traditional	Cloud
Email services	Provide email services by installing Microsoft Exchange on a local server computer.	Contract with an internet-based email provider, such as Gmail (from Google) or Exchange Online (from Microsoft).
Disk storage	Set up a local file server computer with a large amount of shared disk space.	Sign up for Microsoft Office and store your files in Microsoft OneDrive or Microsoft Teams. Or, use Google Drive to store your files in the cloud.
Accounting services	Purchase expensive accounting software and install it on a local server computer.	Sign up for a web-based accounting service.

Looking at the Benefits of Cloud Computing

Cloud computing is a different — and, in many ways, better — approach to networking. The following sections cover a few of the main benefits of moving to cloud-based networking.

Cost

Cloud-based computing typically is less expensive than traditional computing. Consider a typical file server application: To implement a file server, first you have to purchase a file server computer with enough disk space to accommodate your users' needs, perhaps as much as 10TB of disk storage or more. You want the most reliable data storage possible, so you purchase a server-quality computer and fully redundant solid-state disk drives. For the sake of this discussion, figure that the total price of the server — including the disk storage, the operating system license, and the labor cost of setting it up — is about $10,000. Assuming that the server will last for four years, that totals about $2,500 per year.

If you instead acquire your disk storage from a cloud-based file-sharing service, you can expect to pay about one-fourth of that amount for an equivalent amount of storage.

The same economies apply to most other cloud-based solutions. Cloud-based email solutions, for example, typically cost around $5 per month per user — far less than the cost of setting up and maintaining an on-premises Microsoft Exchange Server.

Scalability

So, what happens if you guess wrong about the storage requirements of your file server, and your users end up needing 20TB instead of 10TB? With a traditional file server, you must purchase additional disk drives to accommodate the extra space. Sooner than you want, you'll run out of capacity in the server's cabinet. Then you'll have to purchase an external storage cabinet. Eventually, you'll fill that up, too.

Now suppose that after you expand your server capacity to 20TB, your users' needs contract to just 10TB. Unfortunately, you can't return disk drives for a refund.

REMEMBER

With cloud computing, you pay only for the capacity you're actually using, and you can add capacity whenever you need it. In the file server example, you can write as much data as you need to the cloud storage. Each month, you're billed according to your actual usage. Thus, you don't have to purchase and install additional disk drives to add storage capacity.

Reliability

Especially for smaller businesses, cloud services are much more reliable than in-house services. Just a week before I wrote this chapter, the tape drive that a friend uses to back up his company's data failed. As a result, he was unable to back up data for three days while the tape drive was repaired. Had he been using cloud-based backup, he could've restored his data immediately and wouldn't have been without backups for those four days.

The reason for the increased reliability of cloud services is simply a matter of scale. Most small businesses can't afford the redundancies needed to make their computer operations as reliable as possible. My friend's company can't afford to buy two tape drives so that an extra is available in case the main one fails.

By contrast, cloud services are usually provided by large companies such as Amazon, Google, and Microsoft. These companies have state-of-the-art data centers filled with redundancy. Cloud storage may be kept on multiple servers so that if one server fails, others can take over the load. In some cases, these servers are in different data centers in different parts of the country. Thus, your data will still be available even in the event of a disaster that shuts down an entire data center.

Accessibility

One of the best things about cloud services is that they're available anywhere you have an internet connection. Suppose that you have offices in five cities. Using traditional computing, each office would require its own servers, and you'd have to carefully design systems that allowed users in each of the offices to access shared data.

With cloud computing, each office simply connects to the internet to access the cloud applications. Cloud-based applications are also great if your users are mobile because they can access the applications anywhere they can find an internet connection.

Free of hassles

IT can be a hassle. With cloud-based services, you can outsource the job of complex system maintenance chores, such as upgrades, patches, hardware maintenance, backups, and so on. You get to consume the services while someone else takes care of making sure that the services run properly.

Detailing the Drawbacks of Cloud Computing

Although cloud computing has many advantages over traditional techniques, it isn't without its drawbacks. The following sections outline some of the most significant roadblocks to adopting cloud computing.

Entrenched applications

Your organization may depend on entrenched applications that don't lend themselves especially well to cloud computing — or that at least require significant conversion efforts to migrate to the cloud. For example, you may use an accounting system that relies on local file storage.

Fortunately, many cloud providers offer assistance with this migration. And in many cases, the same application that you run locally can be run in the cloud, so no conversion is necessary.

Internet connection speed

Cloud computing shifts much of the burden of your network to your internet connection. Your users used to access their data on local file servers over gigabit-speed connections, and each user had a relatively dedicated network path to those servers. With cloud computing, everyone has to access data over a single internet connection, often slower than the individual pathways within your local network.

REMEMBER

Although you can upgrade your connection to higher speeds, doing so will cost money — money that may well offset the money you would otherwise have saved by migrating to the cloud.

Internet connection reliability

The cloud resources you access may feature all the redundancy in the world, but if your users access the cloud through a single internet connection, that connection becomes a single point of vulnerability. If it fails, everything that depends on it will fail. Your business may come to a halt until the connection is restored.

TIP

Here are two ways to mitigate this risk:

- **Make sure that you have an enterprise-class internet connection.** Enterprise-class connections are more expensive but provide much better fault tolerance and repair service than consumer-class connections do.
- **Provide redundant connections if you can.** That way, if one connection fails, traffic can be rerouted through alternative connections.

Security threats

You can bet your life that hackers around the world are continually probing for ways to break through the security perimeter of all the major cloud providers. When they do, your data may be exposed.

REMEMBER

Always ensure that strong password policies are enforced!

Examining Three Basic Kinds of Cloud Services

Three distinct kinds of services can be provided via the cloud: applications, platforms, and services (infrastructure). The following sections describe these three types of cloud services in greater detail.

Applications

Most often referred to as *Software as a Service* (SaaS), fully functional applications can be delivered via the cloud. One of the best-known examples is Microsoft 365, which includes cloud-based email (Exchange Online), cloud storage (OneDrive), cloud-based collaboration (Teams), and traditional Office applications (Outlook, Word, Excel, PowerPoint, Access, and more).

A popular alternative to Microsoft 365 is Google Workspace (formerly known as G Suite). Google Workspace provides its own office applications that compete with Microsoft's traditional office applications, as well as cloud-based email and storage.

When you use a cloud-based app, you don't have to worry about any of the details that are commonly associated with running an application on your network, such as deploying the app and applying product upgrades and software patches. Cloud-based apps usually charge a small monthly fee based on the number of users running the software, so costs are low.

Also, as a cloud-based app user, you don't have to worry about providing the hardware or operating system platform on which the application will run. The app provider takes care of those details for you, so you can focus simply on deploying the app to best serve your users' needs.

Platforms

Also referred to as *Platform as a Service* (PaaS), this class of service refers to providers that give you access to a remote virtual operating platform on which you can build your own applications.

At the simplest level, a PaaS provider gives you a complete, functional remote virtual machine (VM) that's fully configured and ready for you to deploy your applications to. If you use a web provider to host your company's website, you're already using PaaS: Most web host providers give you a functioning Linux system, fully configured with all the necessary servers, such as Apache or MySQL. All you have to do is build and deploy your web application on the provider's server.

More-complex PaaS solutions include specialized software that your custom applications can tap to provide services such as data storage, online order processing, and credit card payments.

You have many cloud platform providers to choose from. The best known are AWS, Google Cloud, and Microsoft Azure. Azure is described in Book 5, Chapter 3, and AWS is described in Book 5, Chapter 4.

REMEMBER

When you use PaaS, you take on the responsibility of developing your own custom apps to run on the remote platform. The PaaS provider takes care of the details of maintaining the platform itself, including the base operating system and the hardware on which the platform runs.

Infrastructure

Infrastructure as a Service (IaaS) lets you retain control over your virtual servers but delegate control of the hardware platform on which they run. When you use IaaS, you're purchasing raw computing power that's accessible via the cloud. Typically, IaaS provides you access to remotely hosted VMs, which you can configure however you want.

Public Clouds versus Private Clouds

The most common form of cloud computing uses what is known as a *public cloud* (cloud services that are available to anyone in the world via the internet). Google Workspace is an excellent example of a public cloud service. Anyone with access to the internet can access the public cloud services of Google Workspace: Just point your browser to `https://workspace.google.com` to get started.

A public cloud is like a public utility, in that anyone can subscribe to it on a pay-as-you-go basis. One drawback of public cloud services is that when you use a public cloud service, you entrust your valuable data to a third party that you can't control. You can protect access by using strong passwords, but if your usernames and passwords are compromised, public cloud services can be hacked into and your data can be stolen. Every so often, we all hear news stories about how this company's or that company's backdoor security has been compromised.

A *private cloud* mimics many of the features of cloud computing but is implemented on private hardware within a local network, so it isn't accessible to the general public. Private clouds are inherently more secure because the general public can't access them. Also, they're dependent only on private network connections, so they aren't subject to the limits of a public internet connection.

TIP

As a rule, private clouds are implemented by large organizations that have the resources available to create and maintain their own cloud servers.

A compromise between a public and a private cloud is a *hybrid cloud*, which combines the features of both. Typically, a hybrid cloud system uses a small private cloud for local access to some applications and a public cloud for others. For example, you might maintain your most frequently used data on a private cloud and use the public cloud to store archive data or other less frequently used data.

Introducing Some of the Major Cloud Providers

Although hundreds (even thousands) of companies provide cloud services, most cloud computing is provided by just a few providers.

Amazon

By far the largest provider of cloud services in the world is Amazon. Amazon launched its cloud platform, AWS, in 2006. Since then, hundreds of thousands of customers have signed up. Some of the most notable users of AWS include Airbnb, Netflix, and Pinterest.

AWS includes the following features:

- **Amazon CloudFront:** A PaaS content-delivery system designed to deliver web content to large numbers of users
- **Amazon Elastic Compute Cloud (EC2):** An IaaS system that provides access to raw computing power
- **Amazon Simple Storage Service (S3):** Provides web-based data storage for unlimited amounts of data
- **Amazon Relational Database Service (RDS):** Lets you house your databases in the cloud
- **Amazon Virtual Private Cloud (VPC):** Uses virtual private network (VPN) connections to connect your local network to Amazon's cloud services

For more information about AWS, refer to Book 5, Chapter 4.

Google

Google is also one of the largest providers of cloud services. Its offerings include the following:

- **Google Workspace:** An alternative to Microsoft 365 that provides email, word processing, and spreadsheet functions via the cloud, as well as storage.

Monthly subscription prices range from $7.20 to $21.60 per month, depending on the level of services you need.

- **Google App Engine:** A PaaS interface that lets you develop your own applications that work with Google's cloud services.
- **Google Cloud:** A rich environment in which you can create virtual servers, networks, and storage resources within Google's massive cloud data centers.

Microsoft

Microsoft has its own cloud strategy, designed in part to protect its core business of operating systems and Office applications against competition from other cloud providers, such as Google.

Here are two of Microsoft's cloud offerings:

- **Microsoft 365:** A cloud-based version of Office, which includes Outlook, Teams, SharePoint, Word, Excel, PowerPoint, Access, OneDrive, and many other powerful tools
- **Azure:** A PaaS offering that lets you build websites, deploy VMs that run Windows Server or Linux, or access cloud versions of server applications such as Microsoft SQL Server

For more information about Azure, refer to Book 5, Chapter 3.

Getting into the Cloud

TIP

After you wrap your head around just how cool cloud computing can be, what should you do to move your network toward the cloud? Allow me to make a few recommendations:

- **Don't depend on a poor internet connection.** First and foremost, before you take any of your network operations to the cloud, make sure that you're *not* dependent on a consumer-grade internet connection. Consumer-grade internet connections can be fast, but when an outage occurs, there's no telling how long you'll wait for the connection to be repaired. Invest in a high-speed enterprise-class connection that can scale as your dependence on it increases.

- **Assess which applications you may already have running on the cloud.** If you use Gmail rather than Exchange for your email, congratulations! You've already embraced the cloud. Other examples of cloud services that you may already be using include a remote web host or a file-sharing application such as Dropbox or ShareFile.
- **Don't move to the cloud all at once.** Start by identifying a single application that lends itself to the cloud. For example, if your company archives files when they reach a certain age, look to the cloud for a file storage service.
- **Go with a reputable company.** Amazon, Google, and Microsoft are all huge companies with proven track records in cloud computing. Many other large and established companies also offer cloud services. Don't stake your company's future on a company that didn't exist six months ago.
- **Research, research, research.** Pour yourself into the web, and buy a few books. *Cloud Computing For Dummies,* 2nd Edition, by Daniel Kirsch and Judith Hurwitz (John Wiley & Sons), is a good place to start.

2 Understanding Network Protocols

Contents at a Glance